

BIKKINA ABHIRAM CHOUDHARY

Entry-Level Embedded Systems Engineer | B.Tech ECE Graduate (2026)

Bargarh, India | +91 94398 95696 | abhiramchoudhary005@gmail.com | [linkedin.com/in/abhiram-choudhary-](https://www.linkedin.com/in/abhiram-choudhary-) | github.com/Abhiram23-magnus

SUMMARY

B.Tech graduate in Electronics & Communication Engineering (KL University, May 2026) and Embedded Systems fresher. Develops Embedded C firmware on ARM Cortex-M (STM32) and ESP32 microcontrollers: register-level GPIO, UART, timers/PWM, interrupts (ISR, NVIC) and ADC sensor acquisition, applied in hands-on projects covering RTOS task scheduling, ECG signal acquisition and Wi-Fi video streaming. Debugs firmware on hardware with ST-Link and a logic analyzer; targeting Embedded Software and Firmware Engineer roles.

TECHNICAL SKILLS

- **Programming:** C, Embedded C, C++, Python
- **Embedded Systems:** Firmware development, register-level programming, bare-metal, interrupts (ISR, NVIC), GPIO, timers, PWM, ADC, modular C
- **Microcontrollers:** STM32 (ARM Cortex-M), ESP32, ESP32-CAM
- **Protocols:** UART, SPI, I2C
- **RTOS:** FreeRTOS (tasks, queues, semaphores, mutexes), task scheduling, task priorities, task states
- **Tools & IDEs:** STM32CubeIDE, STM32CubeMX, Keil µVision, Arduino IDE, Git/GitHub
- **Debugging/Development:** ST-Link, logic analyzer, Keil µVision simulator, register-level debugging
- **IoT/Cloud:** Wi-Fi, HTTP server, ESP32 networking

PROJECTS

Mini RTOS Scheduler Simulator | Embedded C | github.com/Abhiram23-magnus/RTOS-Scheduler

- Developed a modular Embedded C simulator modeling RTOS task creation, task states and priority-based scheduling driven by a simulated system tick.
- Implemented task control blocks and queue management using structures and pointers, with each task's routine bound through function pointers.
- Designed scheduler logic that selects the next task from task priority and state on every tick, showing how real-time schedulers make decisions.
- Organized the code into separate source and header modules for tasks, queues and the scheduler, keeping interfaces small and each module testable on its own.
- Documented the design and sample scheduling output on GitHub; built as a learning simulator of RTOS scheduling, not a kernel running on hardware.

ECG Acquisition Firmware with PQRST Analysis | ESP32, AD8232, Embedded C/C++, ADC, Python

Jan 2025 – Feb 2025

- Built single-lead ECG acquisition firmware on ESP32, sampling an AD8232 analog front-end through the on-chip ADC at a fixed sampling rate.
- Implemented signal filtering and buffering in firmware and streamed ECG samples over Wi-Fi to a host application.
- Developed a Python host tool that detects R-peaks, segments PQRST waves, extracts RR-interval and heart-rate features, and classifies heart rhythm with an SVM classifier.
- Documented the system design and results in an IEEE-format research paper.

ESP32-CAM Wireless Surveillance System | ESP32-CAM, OV2640, Embedded C, HTTP

Oct 2024 – Dec 2024

- Built a wireless camera on ESP32-CAM with an OV2640 sensor, streaming JPEG video to a browser through an on-board HTTP server.
- Implemented frame-differencing motion detection in firmware so images are captured only on motion, instead of continuous capture.
- Configured camera resolution and frame buffers in firmware to keep the live stream stable over local Wi-Fi.

INTERNSHIP EXPERIENCE

Embedded Systems Intern — InternzLearn

Aug 2024 – Sep 2024

- Developed bare-metal Embedded C firmware for ARM Cortex-M microcontrollers: GPIO control, timer-based PWM, UART communication and interrupt service routines at register level, without HAL.
- Configured NVIC interrupt priorities and verified ISR timing by toggling GPIO pins and capturing them on a logic analyzer.
- Tested firmware logic in the Keil µVision simulator before flashing, then debugged it on target hardware.

EDUCATION

B.Tech, Electronics & Communication Engineering — KL University (KLEF), Guntur

Aug 2023 – May 2026

CGPA 8.3/10 (lateral entry) | Coursework: Microprocessors & Microcontrollers, Embedded Systems, DSP, Computer Architecture

Diploma, ECE — Vikash Polytechnic College, Bargarh | 72.5%

Jun 2020 – Jun 2023

CERTIFICATIONS

- Embedded Systems Training — Emertxe Information Technologies, Bengaluru, Jul 2026 – Present
- IIoT Programming & Automation — Tessolve Semiconductors, Apr 2025
- NI Certified LabVIEW Associate Developer (CLAD) — National Instruments, Feb 2024